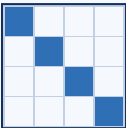


Reflections in the plane

Four mirrors, four matrices. The first three are special cases of the fourth — substituting $\alpha = 0$, and into the general form recovers them exactly.

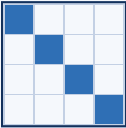
MIRRORS ALONG THE AXES



The x-axis | $\alpha = 0$

MATRIX	EFFECT ON (X, Y)	FIXED DIRECTION
$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	$(x,y) \mapsto (x,-y)$	the x-axis

Diagonal, so its action is entrywise: the first coordinate survives and the second flips. The eigenvalues sit on the diagonal, +1 and -1, exactly as the general result predicts.

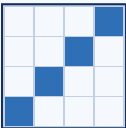


The y-axis | $\alpha = \pi$

MATRIX	EFFECT ON (X, Y)	FIXED DIRECTION
$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$	$(x,y) \mapsto (-x,y)$	the y-axis

The same matrix with the signs exchanged. Note this is not the negative of the row above — that would be $-I$, a rotation by π , which reverses both coordinates and has $\det = +1$.

MIRROR AT 45 DEGREES

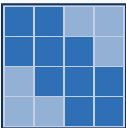


The line $y = x$ | $\alpha = \pi/4$

MATRIX	EFFECT ON (X, Y)	FIXED DIRECTION
$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$(x,y) \mapsto (y,x)$	the line $y = x$

Swapping coordinates is a reflection, which is worth noticing because it also describes the **transpose** at the level of a 2×2 permutation. Its own inverse, as every reflection is.

THE GENERAL MIRROR



Line at angle α | any line through the origin

MATRIX	EFFECT ON (X, Y)	FIXED DIRECTION
$\begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{bmatrix}$	reflects across that line	the mirror line itself

The 2α is no typo: rotating the mirror by α turns the image by 2α , since the incoming and outgoing rays each make angle α with it. Symmetric and orthogonal at once, which is unusual and is why $A^{-1} = A^T = A$.

Every reflection has $\det = -1$, eigenvalues +1 along the mirror and -1 across it, and equals its own inverse. Those three facts hold whatever α is, which is what makes reflection a **family** rather than a list — and why the general row is the only one that has to be remembered.